SDG Blockchain Accelerator Cohort 2
Deliverables Roadmap

Accelerator Duration: 29 September - 15 December 2025
Challenge Owner (CO): UNDP ANGOLA
Solution Maker (SM): GREEN GIRAFFE TECHNOLOGIES

1. Challenge Definition

1.1 Refined Challenge Statement

Despite gains in agricultural productivity driven by initiatives such as UNDP's Kurima Project, many smallholder farmers in rural Angola remain unable to sell their produce at fair and profitable prices. While access to improved inputs, energy, and training has increased yields, structural market barriers continue to lock farmers out of formal and higher-value markets.

These barriers include geographic isolation, high post-harvest losses, limited aggregation and storage capacity, and a heavy dependence on informal middlemen who offer low and opaque pricing. The absence of reliable traceability and quality verification further prevents buyers from confidently sourcing from smallholders, reinforcing mistrust and market exclusion. As a result, farmers—particularly women-led cooperatives—capture little value from their production gains, perpetuating rural poverty and economic vulnerability.

This challenge is fundamentally one of missing market infrastructure. There is no trusted, interoperable digital system that links farmers, aggregators, quality assurance agents, transporters, and buyers through verifiable proof of origin, quality, compliance, and sustainability. Without such infrastructure, productivity gains cannot translate into improved incomes, resilient value chains, or inclusive agricultural growth.

This project directly addresses that gap by enabling transparent, traceable, and trustworthy market connections—transforming smallholder agriculture from high-risk and informal to credible, investable, and market-ready.

1.2 Local Context

This challenge is particularly acute in rural and off-grid provinces such as Huíla, Benguela, Uíge, and Lunda-Norte, where agriculture remains the primary livelihood but market systems are weak and fragmented. In these regions, smallholder farmers—many organized in women-led cooperatives—have begun to improve productivity through access to Kurima-supported agricultural inputs, decentralized energy solutions, and basic agronomic training. As a result, yields have increased and farming households are producing surplus beyond subsistence levels.

However, these productivity gains are not translating into improved incomes or long-term resilience. Post-harvest losses exceed 30%, driven by inadequate storage, limited aggregation infrastructure, unreliable transport, and the absence of structured buyers. Farmers are often forced to sell at the farm gate to informal traders at depressed prices, with little transparency, no quality differentiation, and no bargaining power. This situation disproportionately affects women farmers, who face additional constraints in accessing finance, transport, and market information.

The problem is compounded by limited digital connectivity and the absence of trusted, traceable market linkages. Without reliable systems to verify production, quality, and volumes, farmers struggle to access formal buyers such as processors, retailers, or institutional off-takers. At the same time, buyers perceive smallholder supply as risky and inconsistent, reinforcing reliance on imports and large-scale producers. This trust deficit locks smallholders out of higher-value markets and prevents the emergence of inclusive, competitive value chains.

The broader consequences are significant. Persistently low rural incomes entrench poverty, drive rural-urban migration—particularly among youth—and weaken social cohesion in agricultural communities. At a national level, these inefficiencies undermine Angola's food security, increase dependence on food imports, and slow progress toward economic diversification, a stated priority under the National Development Plan. Agriculture's potential to act as a driver of employment, women's economic empowerment, and climate-resilient growth remains largely unrealized.

While government programs, NGOs, and private actors—including Kurima—are investing in inputs, training, and energy access, market access remains the missing link. Addressing this challenge requires coordinated action across the value chain: farmers and cooperatives producing quality crops; aggregators and transporters reducing losses; quality assurance and extension services building trust and standards; buyers committing to transparent procurement; and digital platforms enabling traceability, pricing transparency, and accountability.

Solving this challenge is therefore not only about improving farmer incomes—it is central to building resilient rural economies, strengthening national food systems, and ensuring that productivity gains translate into inclusive and sustainable development for Angola's most vulnerable agricultural communities.

1.3 Alignment with UNDP Country Office and National Development Priorities

The proposed solution aligns strongly with the UNDP Angola Country Programme (CPD) and the Angola National Development Plan (PND) by addressing structural constraints in rural agriculture through digital innovation, market inclusion, and climate-resilient value chain development.

- ***Economic Diversification & Poverty Reduction***

  Agriculture is a cornerstone of Angola's economic diversification agenda. By enabling smallholder farmers to access formal and premium markets through trusted traceability, quality verification, and transparent transactions, the project increases

farm incomes and reduces reliance on informal intermediaries. This directly supports UNDP's objective of promoting inclusive growth and reducing rural poverty, particularly among women-led cooperatives and vulnerable households.

- ***Digital Transformation & Financial Inclusion***

  The project advances Angola's digital transformation goals by introducing interoperable digital infrastructure in off-grid and rural contexts. By creating verifiable production, quality, and transaction records, it lays the foundation for financial inclusion—enabling farmers and aggregators to build digital identities, access formal buyers, and become visible to financial institutions, insurers, and development programs. This complements UNDP-supported digital public goods and e-governance efforts aimed at strengthening rural service delivery.

- ***Climate Resilience & Sustainable Agriculture***

  The solution integrates environmental and climate-smart data into crop management, quality assurance, and lifecycle assessment processes. This supports national and UNDP priorities on climate resilience by promoting sustainable land use, reducing post-harvest losses, improving resource efficiency, and enabling measurement of environmental performance across the value chain. These capabilities are particularly relevant for adaptation-focused agriculture programs in climate-vulnerable regions.

- ***Youth Employment & Skills Development***

  By formalizing roles such as QA agents, digital aggregators, logistics coordinators, and data-enabled extension services, the project creates new employment pathways for rural youth. These roles align with UNDP's focus on youth entrepreneurship, skills development, and decent work in emerging green and digital economies.

- ***Synergies with UNDP and Donor-Funded Initiatives***

  The project directly complements initiatives such as UNDP's Kurima Project by ensuring that productivity gains translate into market access and income growth. It acts as a downstream market and traceability layer for existing investments in inputs, energy access, and farmer training. Additionally, the solution is designed to interoperate with government, donor, and private-sector programs focused on agricultural commercialization, renewable energy-enabled livelihoods, and value chain strengthening.

1.4 Relevance to UNDP Programmes and Resource Mapping

This pilot is ***not a standalone initiative****; it builds directly on the existing* ***UNDP Kurima Project****, which has already laid critical foundations for rural productivity and resilience in Angola. Kurima has established* ***energy access solutions, productive assets, farmer***

training programs, and Rural Transformation Centers (RTCs) *across multiple provinces, creating the supply-side conditions for increased agricultural output.*

The proposed pilot represents the ***next phase of impact****, addressing a well-documented gap in Kurima's current value chain: the* ***demand-side challenge of market access****. While farmers are now able to produce more, they remain constrained by limited connectivity to formal buyers, lack of trusted quality verification, and absence of traceable transaction infrastructure. Green Giraffe's solution directly responds to this gap by embedding* ***digital market access, quality assurance workflows, escrow-enabled digital payments, and blockchain-based traceability*** *into the existing Kurima ecosystem.*

Scope and Scale
The pilot will be implemented across a selected Kurima-supported Rural Transformation Centers, focusing one farmer group with a target of onboarding and training 50 - 100 farmers focusing on one priority crop already under production. It will engage multiple value chain actors—including farmers, aggregators, QA agents, transporters, processors, and buyers—to demonstrate a full ***farm-to-market traceability and transaction flow****. The initial scope is designed to be modular and scalable, enabling replication across additional provinces and value chains as Kurima expands.*

Timeline
The pilot will run over a ***6 month period****, structured in phases:*

- ***Phase 1:*** *System configuration, stakeholder onboarding, and integration with existing Kurima operations*

- ***Phase 2:*** *Live pilot implementation covering production cycles, quality verification, and market transactions*

- ***Phase 3:*** *Evaluation, learning, and preparation for scale-up across additional RTCs and value chains*

Co-Financing and In-Kind Contributions
The pilot leverages substantial ***in-kind and complementary contributions*** *from existing partners, including:*

- ***Government entities*** *providing infrastructure, policy alignment, and institutional support*

- ***UN agencies*** *such as* ***FAO and UNHCR****, contributing technical expertise, field presence, and programmatic alignment*

- ***NGO partners*** *including* ***ADRA, PIN, and World Vision****, supporting farmer mobilization, training, and community engagement*

Green Giraffe contributes its ***existing technology platform, engineering capacity, and prior R&D investments****, significantly reducing implementation risk and capital*

requirements. This blended approach ensures high leverage of donor resources while accelerating the transition from productivity gains to ***sustainable, market-driven rural incomes****.*

1.5 Key Stakeholders & Potential Partners

The project brings together a multi-stakeholder partnership spanning government, UN agencies, civil society, the private sector, and rural communities. Key stakeholders include:

- ***UNDP Angola*** *– Already onboarded. Serves as the convening partner and lead on programmatic alignment, implementation oversight, and monitoring and evaluation. Pilot-specific coordination will be aligned through UNDP once funding is secured.*

- ***Government of Angola*** *(Ministry of Agriculture and Forestry; Ministry of Energy and Water; Ministry of Trade and Industry) – Already onboarded under the Kurima project. Provides policy alignment, institutional support, and integration with national development priorities. Formal engagement on the pilot will occur once funding is confirmed, within the existing partnership structure.*

- ***Provincial and Local Governments*** *(Huíla and Benguela) – Already onboarded. Facilitate local coordination, access to Rural Transformation Centers, and engagement with farmer groups. Pilot activities will leverage these existing relationships.*

- ***Non-Governmental Organizations (NGOs) and Implementing Partners*** *(ADRA, PIN, SEIVA, World Vision, ADPP) – Already onboarded. Support farmer mobilization, capacity building, extension services, and community engagement. Their involvement in the pilot will build on current Kurima implementation roles.*

- ***Smallholder Farmers and Cooperatives*** *(including women- and youth-led groups) – Already engaged as beneficiaries under Kurima. Will participate in pilot testing related to production, verification, and platform usability.*

- ***Private-Sector Buyers and Processors*** *– Engagement ongoing under Kurima. Participation in the pilot will focus on testing market linkage, quality requirements, and price discovery mechanisms.*

- ***Financial Institutions and Payment Partners*** *(Stanbic Bank; Multicaixa Express) – New onboarding specific to the pilot. Provide settlement, escrow, and digital payment services. Pilot usage will occur within existing agreements.*

- ***Technology and Solution Partner – Green Giraffe*** *– New onboarding specific to the pilot. Will provide the digital traceability and verification solution, working within the established Kurima partnership and governance framework.*

Together, these partners ensure that the solution is institutionally aligned, financially trusted, digitally enabled, and scalable within Angola's agricultural and rural development ecosystem.

1.6 Expected Impact

A successful pilot will demonstrate how trusted digital infrastructure can unlock inclusive, resilient, and market-oriented agricultural value chains in rural Angola. Specifically, the pilot will enable:

- ***Improved farmer incomes and livelihoods*** *through reduced post-harvest losses, fair pricing, and direct access to formal and premium markets.*

- ***Transparent and verifiable market linkages****, allowing buyers, regulators, and development partners to trust data on origin, quality, sustainability, and compliance.*

- ***Strengthened rural service delivery****, by integrating energy access, digital payments, quality assurance, and logistics into a single, coordinated system.*

- ***Empowerment of women and youth****, particularly within cooperatives, by formalizing their participation in value chains and improving access to finance and buyers.*

- ***Systems-level innovation****, establishing a scalable digital public infrastructure model that can be replicated across regions, crops, and development programs.*

- ***Evidence-based policy design****, by generating reliable, real-time data on production, losses, pricing, and compliance to inform agricultural, trade, and food security policies.*

- ***A pathway to scale****, enabling government and UNDP to transition from pilot implementation to national adoption, private-sector engagement, and blended financing models.*

From the Country Office's perspective, success is not only measured by pilot performance, but by the creation of a durable, interoperable system that strengthens institutional capacity, supports national development priorities, and accelerates inclusive rural economic transformation.

1.7 Target SDGs and Indicators

The challenge and proposed solution will directly contribute to the following SDG indicators:

- ***SDG 1.4.2 – Proportion of adults with secure rights to economic resources, including access to financial services***
  The project enables smallholder farmers to access formal markets, digital payments, and escrow-based settlement systems, strengthening their financial inclusion and

control over income from agricultural production.

- ***SDG 2.3.2 – Average income of small-scale food producers, by sex and indigenous status***
  By reducing post-harvest losses, improving price transparency, and connecting farmers to verified buyers, the solution directly increases net incomes for small-scale producers, with particular benefits for women-led cooperatives.

- ***SDG 9.c.1 – Proportion of population covered by a mobile network, by technology***
  The pilot expands the use of digital infrastructure in rural and off-grid areas by embedding mobile-enabled data capture, payments, and traceability into agricultural value chains, supporting inclusive digital transformation.

2. User Mapping & Testing Strategy

2.1 Primary User Persona

The primary users of the solution are smallholder farmers, exemplified by Walisa, a woman farmer living in rural Huíla Province. Walisa cultivates maize, groundnuts, and vegetables on less than two hectares of land. Like many farmers in her community, she relies on rain-fed agriculture, has limited access to formal markets, and often sells produce to informal traders at low and unpredictable prices.

Despite participating in productivity-enhancing initiatives such as improved inputs, training, and energy access, Walisa faces persistent challenges after harvest—post-harvest losses, lack of storage, delayed or unfair payments, and no verifiable proof of the quality or origin of her produce. She also has limited access to digital financial services and market information.

This solution is designed to serve farmers like Walisa by enabling trusted market access, transparent pricing, secure digital payments, and verifiable traceability, allowing her to sell her produce with confidence, receive fair value, and invest sustainably in her farm and household

2.2 User Story

Primary User – Smallholder Farmer

- *As a smallholder farmer, I want to sell my produce through a trusted digital system, so that I receive fair prices, timely payment, and recognition for the quality and sustainability of my crops.*

Secondary Users

QA Agent

- *As a QA agent, I want to verify farms, batches, and compliance through geo-tagged inspections, so that only credible, compliant produce enters formal markets.*

Aggregator / Cooperative

- *As an aggregator, I want to consolidate produce from verified farmers with clear quality and quantity records, so that I can supply buyers reliably and reduce disputes.*

Buyer / Processor

- *As a buyer, I want to purchase traceable and verified agricultural products, so that I can meet quality, sustainability, and regulatory requirements with confidence.*

System / Program Admin (UNDP / Government)

- *As a program administrator, I want visibility into verified value chains and outcomes, so that I can monitor impact, inform policy, and scale successful rural development models.*

2.3 Testing Strategy

User testing will be conducted in phased, real-world pilots embedded within active agricultural value chains. Green Giraffe has already completed an initial pilot using its own fully traceable snack products, validating core functionalities such as farmer onboarding, batch creation, traceability, and basic QA verification.

*A second pilot is scheduled with **260Brands**, an organic soya bean processor in Zambia. This pilot will test end-to-end workflows including farmer aggregation, QA verification, batch reconciliation, buyer requirements, and digital settlement under live commercial conditions.*

Testing will involve:

- ***Smallholder farmers** testing usability of onboarding, batch reporting, and payment receipt*

- ***QA agents** testing inspection, verification, reconciliation, and geo-tagged visitation workflows*

- ***Processors and buyers** testing traceability visibility, compliance reports, and batch acceptance*

- ***System administrators** testing project setup, reporting, and audit readiness*

User feedback will be collected through observation during use, structured interviews, and system analytics. Insights will be used to iteratively refine user experience, workflows, and data models before scaling to additional value chains and regions.

2.4 Feedback Loop Design

Feedback will be gathered continuously from users and stakeholders throughout the pilot using a combination of qualitative and quantitative methods embedded directly into operations.

Feedback Gathering

- ***In-app prompts and flags:*** *Farmers, QA agents, buyers, and administrators will be able to flag issues, confusion points, or suggestions during normal system use.*

- ***Structured debriefs:*** *Regular check-ins with farmers, QA agents, processors, and buyers will be conducted at key milestones (onboarding, first sale, inspection, payment).*

- ***Field observations:*** *QA agents and project staff will document usability challenges and behavioral insights during trainings, inspections, and farm visits.*

- ***Stakeholder reviews:*** *Periodic reviews with UNDP, government counterparts, and implementing partners will assess alignment with policy goals, safeguards, and operational realities.*

- ***System analytics:*** *Usage data (completion rates, error frequency, time-to-complete workflows) will identify friction points and adoption barriers.*

Feedback Integration

- *Feedback will be logged in a centralized learning backlog and categorized by urgency, user type, and system component.*

- *Priority issues affecting trust, payments, compliance, or usability will be addressed immediately.*

- *Iterative improvements will be deployed in short development cycles, with changes communicated back to users.*

- *Lessons learned will inform updated training materials, system configurations, and pilot design.*

- *Synthesized insights will be shared with UNDP and partners to support evidence-based decisions and potential scale-up.*

This approach ensures the pilot remains user-centered, adaptive, and grounded in real operational learning while generating insights for broader system transformation.

3. Solution Overview

3.1 Solution Summary

At the heart of this system is Cardano blockchain technology, which makes the entire solution trustworthy, auditable, and tamper-proof. The solution begins with a digital demand listing feature, where buyers—processors, aggregators, or institutional off-takers—publish verified crop requirements. Smallholder farmers subscribe to these listings, solving the market access problem by connecting supply directly to verified demand.

Each farmer, farm, and plot receives a digital identity anchored to GPS-mapped land boundaries, which serves as a secure reference for all activities. Crop management logs—including planting, input application, harvest, and storage—are persisted on Cardano, creating an immutable chain of custody that ensures no data can be tampered with.

QA Agents are essential gatekeepers in this system. They verify farm boundaries, crop activities, harvest quantities, and quality standards. Only QA-verified metadata is written to the blockchain, guaranteeing that subsequent steps—reconciliation, payments, and reporting—are based on accurate, trusted data.

Payments are securely linked to verified outputs, ensuring the right farmer is paid for the verified produce, reducing fraud and errors. At the processing stage, the ERP system maintains full traceability across multiple batches and ingredients, with every final product remaining linked to its source, enforcing global food safety and compliance standards (e.g., HACCP, ISO 22000, organic certification).

3.2 Core Features and Functionalities

1. *Demand Listing & Market Access*

- *Buyers post verified crop demand.*
- *Farmers subscribe to meet specific demands, creating direct market linkages.*
- *Solves the problem of unreliable and exploitative middlemen.*

2. Digital Farmer Identity & Mapping

- *Each farmer, farm, and plot has a digital identity.*
- *GPS-mapped fields ensure location verification and traceability.*
- *Forms the backbone for crop management, QA, and payments.*

3. Crop Management & Activity Logging

- *Farmers record crop activities: planting, inputs, pest management, harvesting.*
- *All records are persisted on blockchain, forming an immutable chain of custody.*
- *Enables traceable, evidence-based compliance and reporting.*

4. QA Verification Module

- *QA Agents verify boundaries, crops, harvests, and quality.*
- *Only QA-verified data is written to blockchain.*
- *Ensures compliance with global standards and accurate farmer payments.*
- *Supports farmer training, farm visitation, and continuous monitoring.*

5. Blockchain-Enabled Traceability & Payments

- *Immutable, auditable records linking farmers to batches.*
- *Verified reconciliation ensures the right farmer is paid.*
- *Reduces fraud, errors, and disputes.*
- *Provides foundation for Life Cycle Assessment (LCA) and ESG reporting.*

6. Processing ERP Integration

- *Tracks ingredients across multiple batches back to original farmers.*
- *Maintains global food safety compliance (HACCP, ISO 22000, Organic).*
- *Supports batch-level traceability and supply chain accountability.*

7. Compliance & ESG Reporting

- *Generates QA compliance summaries and ESG metrics.*
- *Supports regulatory reporting, donor requirements, and investor confidence.*
- *Links directly to the LCA dashboard for cradle-to-grave impact analysis.*

8. Analytics & Dashboard

- *Personalized dashboards for QA Agents and Admins.*
- *Tracks farmer performance, training, batch status, and compliance.*
- *Provides actionable alerts and insights for interventions.*

3.3 Tech Stack and Tools

1. Blockchain & Verification

- ***Cardano Blockchain** – Immutable, auditable, decentralized ledger powering traceability, QA verification, and secure financial reconciliation.*

- ***Role:** Ensures trust across the value chain; without it, the system cannot guarantee verified provenance or compliance.*

2. AI & Analytics

- ***Ollama** – AI-driven insights for crop management, QA verification, and ESG analysis.*

- ***Prithvi** – Analysis layer for geospatial and productivity analytics (does not generate raw data).*

3. Life Cycle Assessment

- ***Agribalayse 3.2 LCA** – Cradle-to-grave impact assessment, ESG reporting, and sustainability evaluation.*

4. Remote Sensing & Geospatial Data

- ***NASA HLS (Harmonized Landsat & Sentinel-2)*** *– Satellite imagery for crop monitoring, land use, and verification.*

- ***iSDANASA HLS*** *– Harmonized Landsat & Sentinel-2 dataset integrated for agricultural insights.*

- ***Global Forest Watch (GFW)*** *– Deforestation and land cover monitoring.*

- ***OpenWeatherMap Agro API*** *– Localized weather and climate data for crop planning and risk management.*

- ***iSDAsoil API*** *– Soil quality and fertility analytics.*

5. Software & Development Stack

- ***Programming Languages:*** *Python (AI, analytics, ETL), Java (backend), PHP/Laravel (web application).*

- ***Database:*** *PostgreSQL / MySQL for relational data storage and integration with blockchain metadata.*

- ***APIs & Microservices:*** *RESTful APIs for integration with GIS, LCA, remote sensing, and payment gateways.*

6. DevOps & Cloud Hosting

- ***Cloud Hosting:*** *Railway for deployment of web applications and APIs.*

- ***Storage:*** *Wasabi for immutable file storage (images, documents, batch records).*

- ***CI/CD & DevOps Tools:*** *Docker, GitHub Actions, automated testing, and containerized deployment for reliable updates.*

7. Financial & Payment Integration

- ***Stanbic Bank Zambia:*** *Escrow and settlement banking for verified payments.*

- ***Multicaxia Express:*** *Payment gateway for digital collection and disbursement.*

3.4 Blockchain Functionality

On-Chain Verification & Traceability

- ***Immutable crop & batch records****: Farmer logs (crop management, harvest, batch creation) are hashed and stored on Cardano.*

- ***QA verification****: Only QA-verified data (inspections, compliance, lab results) are anchored on-chain.*

- ***Audit trail**: Every version of a record is blockchain-anchored, providing tamper-proof proof of origin and history.*

Smart Contracts

- ***Reconciliation & Payment Triggers**: Smart contracts release escrowed payments automatically based on verified batch quantities, reconciled weights, and accepted quality grades.*

- ***Conditional Disbursement**: Contracts enforce rules like partial acceptance, downgrades, or full rejection of batches.*

- ***Stakeholder Roles Enforcement**: QA Agents cannot override payments; admins alone execute financial flows, ensuring separation of duties.*

Token Minting & Asset Representation (Optional/Future)

- ***Batch Tokens**: Each batch can be represented as a unique token (NFT-style) to trace provenance, link multiple ingredients, or represent ownership and quality certification.*

- ***Stakeholder Credits**: Optional tokenized incentives for farmers who meet ESG, training, or quality standards.*

Data Hashing & Anchoring

- ***Immutable JSON**: All forms, farmer logs, QA reports, lab results, and training records are hashed before storing on-chain.*

- ***Metadata Anchoring**: IDs, timestamps, GPS coordinates, batch IDs, and QA signatures are included in the blockchain record.*

Compliance Enforcement

- ***On-Chain Evidence**: EU Organic, EUDR, ISO, HACCP, and GMP compliance forms are anchored to prevent tampering.*

- ***Final Verification Lock**: Once a batch is finalized, blockchain ensures no further edits can occur, securing traceability from farm-to-fork.*

4. Weekly Workplan

Onboarding & Setup (Weeks 1–6 | 6 weeks)

- *Define project scope, stakeholders, and geographic coverage.*
- *Configure QA jurisdictions, aggregation points, and farmer enrollments.*
- *Integrate blockchain infrastructure, payment gateways, and ERP systems.*

Demand Listing & Farmer Subscription (Weeks 7–12 | 6 weeks)

- *Publish market demand on the platform.*
- *Facilitate farmer subscriptions to meet demand.*
- *Verify farmer digital identities, maps, and crop management logs.*

Crop Management & QA Verification (Weeks 13–36 | 24 weeks)

- *Monitor farmer crop activities through the digital platform.*
- *Conduct QA inspections, training, and field visits.*
- *Persist only verified data on-chain for traceability.*

Harvest, Batch Creation & Reconciliation (Weeks 37–48 | 12 weeks)

- *Record harvests and batch details.*
- *Reconcile declared vs. inspected quantities.*
- *Lock batch for processing and initiate escrow logic via smart contracts.*

Processing & Traceable Production (Weeks 49–60 | 12 weeks)

- *Manage ERP operations to ensure multi-ingredient traceability.*
- *Maintain compliance with food safety and sustainability standards.*
- *Persist verified processing data on-chain.*

Reporting & Analysis (Weeks 61–72 | 12 weeks)

- *Generate QA Master Reports, LCA reports, and ESG summaries.*
- *Review analytics for farmer performance, training, and compliance gaps.*
- *Provide evidence for policy, scaling, and investment decisions.*

Scaling & Continuous Improvement (Weeks 73–78 | 6 weeks)

- *Identify bottlenecks or gaps in adoption.*
- *Optimize platform workflows, farmer engagement, and QA verification.*
- *Expand to new value chains, regions, or markets while maintaining full traceability.*

5. Final Deliverables

5.1 Solution Demo (PoC)

https://youtu.be/JOa8aQzBHsM

5.2 Testing & Feedback Documentation

User Testing Activities:

- ***Pilot with Green Giraffe snacks*** *– tested full traceability and QA verification workflows.*

- ***Pilot with 260Brands (organic soya processor)*** *– tested demand listing, farmer subscription, and batch reconciliation.*

- ***QA Agent Field Trials*** *– tested farmer onboarding, crop inspection, training logging, and geo-tagged visit verification.*

- ***System Admin Trials*** *– tested project creation, stakeholder management, and LCA report generation.*

Challenges Observed:

- *Some farmers struggled with mobile connectivity and onboarding.*

- *QA Agents initially found multi-tab dashboards complex; required simplified KPIs and drill-down views.*

- *Lab result submission workflows occasionally delayed batch verification.*

Feedback Gathered:

- *Requests for automated alerts for pending inspections and farmer training.*

- *Need for clearer separation of financial vs QA responsibilities.*

- *Desire for faster reconciliation workflows and easier audit trail access.*

5.3 Feedback Integration Summary

Changes Made Based on Feedback:

- *Introduced role-based access control, limiting QA Agents to verification and reconciliation only; financial data is admin-only.*
- *Simplified QA master dashboard into multi-tab view with collapsible tables and color-coded KPIs.*
- *Added automated alerts for farm visits, training gaps, and pending lab results.*
- *Enhanced geo-tagged farmer visitation module to reduce fraud and improve compliance tracking.*
- *Streamlined batch reconciliation forms to speed up processing and ensure accurate escrow payments.*
- *Improved onboarding workflows for farmers in low-connectivity areas.*
- *Versioning and blockchain anchoring reinforced for all compliance, QA, and traceability records.*

5.4 Pilot-Ready Solution

https://greengiraffetechnologies.com/

5.5 Confirmed Pilot Partnership(s)

Pilot Partner: UNDP Kurima Initiative (Angola)

- ***Scope:*** *Leverages existing rural transformation centers, energy access, and farmer support infrastructure across Huíla, Benguela, Uíge, and Lunda-Norte.*
- ***Commitment:*** *Co-implementation of Green Giraffe's blockchain-enabled market access and traceability solution, including QA verification, farmer onboarding, and LCA reporting.*
- ***Additional Support: I****n-kind contributions from UN agencies (FAO, UNHCR) and NGOs (ADRA, PIN, World Vision).*
- ***Role:*** *UNDP Kurima provides farmer networks, technical guidance, and coordination with provincial governments; Green Giraffe provides digital infrastructure, QA and ERP systems, and blockchain traceability.*

Other Stakeholders:

- ***Stanbic Bank & Multicaixa Express:*** *Payment gateway and settlement support for secure financial flows to farmers.*
- ***Provincial Governments & Ministries:*** *Facilitate approvals, farmer registration, and alignment with national development priorities.*
- ***Confirmation Status:*** *Documented agreements exist with UNDP Kurima and key partners, confirming active participation in pilot implementation.*

5.6 Stakeholder Alignment Matrix

Stakeholder / Partner	Role in the Pilot	Key Responsibilities	Level of Involvement	Engagement Modality
UNDP Angola	Challenge Owner	Programme oversight, coordination, policy alignment	Lead	MoU
Green Giraffe Technologies	Solution Maker	Platform development, deployment, analytics, reporting	Lead	MoU
MINAGRIP	Government Partner	Farmer mobilization, extension services, policy support	Support	LoI
Provincial Governments	Local Implementation	Local coordination, facilities, outreach	Support	MoU

(Huíla, Benguela)				
ADRA / PIN / World Vision	Implementing Partners	Farmer training, onboarding, digital literacy	Support	Contract
Private Buyers / Processors	Market Off-takers	Commodity sourcing, pricing, compliance feedback	Advisory	Commercial Agreement
Financial Institutions / Mobile Money	Payments Partner	Instant payments, escrow settlement	Support	API Agreement

5.7 Budget Proposal (Pilot)

Budget Line Item	**Unit / Description**	**Quantity**	**Unit Cost (USD)**	**Total Cost (USD)**	**Funding Source**
Platform Development	AI, blockchain, marketplace modules	1	5,000	5,000	Grant Pending
QA Agent Training	Field agents onboarding & training	5	500	2,500	Grant Pending
Field Devices	Biometric kits, tablets, sensors	5	800	4,000	Grant Pending
Cloud & Infrastructure	Hosting, APIs, data services	12 months	400	4,800	Grant Pending
Farmer Training	Digital literacy & onboarding	100 farmers	25	2,500	Grant Pending
Field Visitation & Farmer Onboarding Travel	Joint field visits (transport, accomodation, fuel, per diem) for Green Giraffe, UNDP Angola, and Kurima Project staff to	6 trips (1 per month for UNDP Angola and Kurima Project staff and 2 trips to Angola from the Green Giraffe team at the start and	31,200 (lump sum)	31,200	Grant Pending

	onboard and train farmers over 6 months	at the end of the pilot)			

TOTAL ESTIMATED PILOT COST: USD 50,000

5.8 Impact Measurement Framework

5.8.1.1 Contribution to SDGs

SDG	Indicator	Targeted Impact
SDG 1.4.2	Access to economic resources and financial services	Enable smallholder farmers to access secure payments, fair pricing, and digital financial services via Stanbic Bank & Multicaixa Express integration.
SDG 2.3.2	Average income of small-scale food producers	Increase farmer income through verified market access, fair trade, and value chain efficiency using blockchain-enabled traceability.
SDG 12.3	Reduction in food loss and waste	Reduce post-harvest losses (>30% currently) via traceable logistics, QA verification, and improved aggregation & storage practices.

5.8.1.2 Impact Tracking Mechanisms

- **Blockchain audit trail:** Immutable records of farmer activities, QA verification, and batch traceability.
- **LCA dashboard (Agribalayse 3.2):** Measures environmental impact, resource use efficiency, and compliance with sustainability standards.
- **Farmer dashboards:** Track training, adoption of best practices, and progress toward yield and quality improvements.
- **Financial reconciliation:** Monitors payments, downgrades, and accepted quantities to ensure fair income distribution.
- **Lab & compliance results:** Measure food safety standards adherence and quality improvements.

5.8.1.3 Frequency of Measurement

- Farmer-level data: **Monthly**
- QA and compliance records: **Per crop cycle / batch**
- Financial flows: **Per purchase / transaction**
- Environmental LCA metrics: **Quarterly**

5.8.2 Key Performance Indicators (KPIs)

KPI	Measurement Method	Target	Notes
Farmers onboarded	Digital IDs issued	1,000	Huíla & Benguela
Reduction in post-harvest loss	Storage & sales data	30%	Perishable crops
Verified transactions	On-chain records	5,000+	Pilot period
Income increase	Baseline vs pilot	15–20%	Farmer income

5.8.3 Monitoring & Reporting Plan

Objective: *Track project progress, measure impact, and ensure transparency and accountability in alignment with SDG targets.*

- ***Frequency:***
    1. ***Monthly:*** *Data collection and preliminary dashboards.*
    2. ***Quarterly:*** *Consolidated reports, including insights, challenges, and recommendations.*
    3. ***Annually:*** *Comprehensive impact assessment aligning with SDGs.*
- ***Responsible Parties:***
    1. ***Green Giraffe:*** *Data collection, initial analysis, field reports.*
    2. ***UNDP Angola:*** *Oversight, verification, strategic reporting, and dissemination to stakeholders.*
- ***Reporting Formats:***

1. ***Dashboards:*** *Visual representation of KPIs and real-time progress.*

2. ***Field Reports:*** *Narrative documentation from on-the-ground activities.*

3. ***Quarterly Reports:*** *Aggregated data with insights, lessons learned, and recommendations.*

- ***Tools & Methods:***

    1. ***Blockchain Analytics:*** *Ensure transparency and traceability of transactions and resource flows.*

    2. ***AI Dashboards:*** *Automated analytics for predictive trends, anomaly detection, and performance insights.*

    3. ***Field Reports:*** *Qualitative and quantitative observations from project sites.*

    4. ***GIS Mapping & Satellite Imagery:*** *Track geospatial impact of interventions.*

- ***Key Performance Indicators (KPIs):***

    1. *Number of farmers adopting sustainable practices.*

    2. *Volume of produce tracked via blockchain.*

    3. *Reduction in food loss/waste percentages.*

    4. *Stakeholder engagement and training completion rates.*

    5. *Compliance with regulatory requirements.*

- ***Workflow:***

    1. *Field data collected weekly via mobile apps.*

    2. *Data uploaded to AI-enabled dashboards.*

    3. *Monthly dashboards generated for internal review.*

    4. *Quarterly field visits and audits for verification.*

    5. *Consolidated reports submitted to UNDP and stakeholders.*

5.8.4 Risk Assessment & Mitigation

Objective: *Identify potential risks and implement strategies to minimize negative impact on project delivery.*

- ***Connectivity Risks:***
    - ***Description:*** *Limited or unstable internet connectivity in rural areas may affect data collection.*
    - ***Mitigation:*** *Offline-first mobile apps with automatic syncing when connectivity is restored.*
- ***Adoption Risks:***
    - ***Description:*** *Low uptake of new technology and practices by farmers or local partners.*
    - ***Mitigation:*** *NGO-led training sessions, peer-to-peer learning networks, and incentivized adoption programs.*
- ***Regulatory Risks:***
    - ***Description:*** *Changes in government policy or compliance requirements may affect operations.*
    - ***Mitigation:*** *Continuous engagement with government authorities, participation in policy dialogues, and legal compliance monitoring.*
- ***Financial Risks:***
    - ***Description:*** *Delays in funding or cash flow issues could slow project implementation.*
    - ***Mitigation:*** *Diversified funding sources, milestone-based budgeting, and contingency reserves.*
- ***Operational Risks:***
    - ***Description:*** *Equipment failure or supply chain disruptions.*
    - ***Mitigation:*** *Regular maintenance, backup suppliers, and inventory buffers.*
- ***Monitoring & Escalation:***
    - *Monthly review of risk indicators via dashboards.*
    - *Immediate reporting of critical risks to project leadership.*
    - *Quarterly risk audit alongside performance reporting.*

5.9 Post-Acceleration Roadmap & Sustainability Plan

5.9.1 Phased Implementation Timeline

This is how the project will evolve beyond the acceleration phase, focusing on phased implementation, financial sustainability, and strategic fundraising to ensure long-term impact and scalability.

Phase	Duration	Key Activities	Targets / Outcomes
Short Term	0–6 months	Pilot execution, field validation, technology testing, stakeholder onboarding	Validate platform functionality, engage initial cohort of farmers, gather user feedback for improvement
Medium Term	6–18 months	National scale-up to 10,000 farmers, partnerships with NGOs, government agencies, and private sector	Expand farmer adoption, establish revenue streams, demonstrate measurable impact on productivity and sustainability
Long Term	18+ months	Regional expansion across Southern Africa, replication in neighboring countries, cross-border partnerships	Strengthen regional presence, influence agricultural digital transformation policies, achieve significant market penetration and environmental impact

5.9.2 Financial Sustainability Model

- *The project will transition from pilot funding to self-sustaining operations through diversified revenue streams:*
- *SaaS Licensing: Subscription fees from agribusinesses, cooperatives, and NGOs using the platform.*
- *API Integrations: Fees from third-party service providers integrating data, analytics, or payment services.*
- *Marketplace Commissions: Transaction fees from farmers and buyers trading produce through the platform.*
- *Certification & Carbon MRV Services: Revenue from auditing, carbon credit certification, and environmental impact reporting for sustainability-conscious buyers and investors.*

Key Considerations:

- *Maintain affordability for smallholder farmers while ensuring operational cost recovery.*
- *Monitor revenue trends and adjust pricing, partnerships, or service tiers to sustain growth.*

5.9.3 Fundraising Plan

- *To support growth and expansion, funding will be sourced from multiple channels:*
- *Multilateral Development Institutions: UNDP, AfDB, EU Blending Facilities.*
- *Climate & Sustainability Funds: Green Climate Fund, national climate initiatives, regional environmental programs.*
- *Strategic Private-Sector Investors: Agri-tech investors, corporate social responsibility programs, and impact investors seeking measurable social and environmental returns.*

Approach:

- *Develop clear investment propositions, highlighting social, economic, and environmental impact.*
- *Leverage pilot data and early adoption metrics to attract funding.*
- *Maintain a balanced funding strategy to reduce dependency on any single source and ensure resilience.*

5.11 Legal & Compliance Considerations

To ensure that the project activities comply with Angola's legal and regulatory environment, the following actions and recommendations will be implemented to protect stakeholders, minimize legal risk, and maintain operational integrity.

1. Data Protection & Privacy

- ***Personal Data Protection Law (Law No. 22/11 of 17 June 2011):***
  Regulates the collection, processing, storage, and transmission of personal data. Guarantees transparency, confidentiality, and individual rights to access, correct, and oppose processing of personal data. Compliance is monitored by the ***Angolan Data Protection Agency (APD)****.*

- ***Complementary Legal Framework:***
  - *Law No. 23/11 on Electronic Communications and Information Society Services.*
  - *Law No. 7/17 on Protection of Information Systems and Networks.*

- ***Compliance Actions:***
  - *Register data processing activities with APD if required.*
  - *Implement encryption, access control, and anonymization where necessary.*

- *Maintain transparent data policies and informed consent procedures.*
- *Monitor changes to draft amendments under public consultation.*

2. Agricultural Laws & Trade Regulations

- *Ensure compliance with national standards for agricultural products, quality, and safety.*
- *Align digital agriculture activities (data collection, tracking, remote sensing) with applicable data protection laws.*
- *Use legally compliant contracts and data sharing agreements with farmers and partners.*

3. Digital Payments & Financial Transactions

- ***Payments System Law (Law No. 40/20 of 16 December 2020):***
  Governs electronic payments, mobile money, and other digital financial instruments under oversight of the ***National Bank of Angola (BNA)****.*
- ***Compliance Actions:***
  - *Verify registration and licensing requirements with BNA.*
  - *Implement anti-money laundering (AML) and counter-terrorist financing (CTF) measures.*
  - *Ensure secure payment systems and data protection.*

4. Blockchain & Digital Asset Regulation

- ***Cryptocurrency & Mining (Law No. 3/24 of 10 April 2024):***
  Mining of cryptocurrencies and related activities is prohibited in Angola.
- ***Blockchain Use for Tracking/Record-Keeping:***
  Permissible if it does not involve prohibited activities. Legal certainty is still developing.
- ***Compliance Actions:***
  - *Avoid cryptocurrency mining or trading.*
  - *Engage regulators if any tokenization or digital asset-related functions are planned.*

5. Cybersecurity & IT Risk Management

- *Angola's National Cybersecurity Strategy strengthens digital infrastructure and protects critical systems.*

- ***Compliance Actions:***

    - *Implement cybersecurity best practices (penetration testing, incident response).*

    - *Ensure project systems meet national cybersecurity requirements.*

Enforcement & Penalties

- ***Data Protection Violations:*** *Fines and administrative sanctions; possible criminal penalties.*

- ***Cryptocurrency Mining Violations:*** *Criminal penalties including fines and imprisonment.*

- ***Digital Payments/AML Violations:*** *Regulatory sanctions, including restrictions on operations.*

Summary of Key Compliance Requirements (Angola)

Legal Area	**Key Requirement**	**Main Authority**
Data Protection	Law No. 22/11 – secure processing & consent	APD
Electronic Communications	Law No. 23/11 – data & digital service rules	MINTTICS / APD
Payments & Digital Finance	Law No. 40/20 – regulated payments system	BNA
Crypto & Digital Assets	Law No. 3/24 – mining prohibited	CMC / BNA
Cybersecurity	National strategy & cybersecurity laws	MINTTICS / APD

5.12 Visibility Assets / Communication Materials

One-pager, impact visuals, farmer stories, and demo videos for UNDP and partners.

5.13 Pitch Deck

A concise pitch deck highlighting problem, solution, impact, and scalability.

https://drive.google.com/file/d/168cTZHoHJCaB1DXSPZNTAVB_31Q2aiSK/view?usp=drivesdk

5.14 Final Demo Presentation

Live demonstration of the platform showing farmer onboarding, traceability, marketplace transaction, and payment flow.

https://www.youtube.com/watch?v=JOa8aQzBHsM&t=15s

Deliverable Completion Table

Please indicate your intended completion week (or date) for each deliverable. You can revise this plan as the project evolves.

Deliverable	Target Completion (Week # or Date)
1. Challenge Definition (incl. local context, SDG contributions, partners, impact)	Completed
2. User Mapping & Testing Strategy (personas, user stories, feedback loop)	Completed
3. Solution Overview (core features, tech stack, blockchain functionality)	Completed
5.1 Solution Demo (PoC)	Completed
5.2 Testing & Feedback Documentation	Completed
5.3 Feedback Integration Summary	Completed

5.4 Pilot-Ready Solution	Completed
5.5 Confirmed Pilot Partnership(s)	Completed
5.6 Stakeholder Alignment Matrix	Completed
5.7 Budget Proposal (Pilot)	Completed
5.8 Impact Measurement Framework (KPIs, SDG indicators, risk plan)	Completed
5.9 Sustainability Plan & Post-Acceleration Roadmap (implementation timeline, budget, partners, fundraising)	Completed
5.11 Legal & Compliance Mapping	Completed
5.12 Visibility Assets / Communication Materials	Completed
5.13 Pitch Deck	Completed